Release Notes

CADP internal QA build release notes — generated from OpenProject. Not the in-app Release Notes feature used by tenants.

1. Release Overview

Field	Details
Product Name	**CADP Workspace**
Release Version	**Release 1.0.0 (25)**
Release Date	**October 23, 2026**
Release Type	**Feature & Bug Fix Release**
Prepared By	**Development Team**
Approved By	**Ganagavathy**
Environment	**QA**

2. Release Summary

This release of CADP Workspace Version 1.0.0 (25) delivers 0 bug fixes, 5 new features, and 16 enhancements across the platform. Highlights include Download PDF for user manuals (same pipeline as Release Notes), Download PDF restored (code was missing behind the two entries above), and Table rows obey their columns' rules: Required, allowed dates, length, pattern, email. It also includes a required configuration change (see Configuration Changes). Refer to the Known Issues section for current limitations.

3. New Features

Feature ID	Feature Name	Description	Impact
	Download PDF for user manuals (same pipeline as Release Notes)	New `GET /user-manual/download-pdf/:userManualId` returns the generated manual as; a PDF attachment. `userManualDocument.service.js` fills the manual into its selected; template using the same fields as the frontend Preview. `buildManualTokens` mirrors; `ManualPreviewTab#buildManualTokens`: the numbered document, the TOC and the per-section; fields. The template is then put through Release Notes' print pipeline:; `upgradeLegacyTemplateHtml` → `substituteTokens` → `normalizeFlowPositioning` →; `buildPrintSafeCss`, then the template's paperSetup/defaultFont → template-service; `renderHtmlToPdf`. With no template (or one that is missing), it uses the standard layout; at A4 with 15mm margins. The file is named `User Manual(<name> v<version>)-DD-MM-YY HH-MM-SS.pdf`,; ASCII-only because Node rejects non-ASCII text in Content-Disposition. A manual that has not; been generated returns 409. Tests: `userManualDocument.test.js` (3).	Requested: user manuals should download like Release Notes.
	Download PDF restored (code was missing behind the two entries above)	The two entries above were logged, but their code was not in the repo, so this; change re-implements them. `GET /user-manual/download-pdf/:userManualId` and; `userManualDocument.service.js`: tokens mirror the Preview, then `upgradeLegacyTemplateHtml`; → `substituteTokens` → `dropBodyPageBreaks` → `normalizeFlowPositioning` → `buildPrintSafeCss`; → template-service `renderHtmlToPdf` at scale 1. Templates saved without page margins now get; 15mm margins in their own unit, so later pages don't print to the paper edge. Headings stay; with their text, images are never split across pages, and the 794px template canvas fits; inside the margins. The backend `upgradeLegacyTemplateHtml` also drops the; "Appendix — Screenshots" heading and the `{{screenshots}}` slot. Tests: `userManualDocument.test.js` (5).	The User Manual Preview downloaded a `.html` file. It now downloads a PDF like Release Notes.

Feature ID	Feature Name	Description	Impact
	Table rows obey their columns' rules: Required, allowed dates, length, pattern, email	**Row checks (definition tables, runtime only).** "+" refuses the entry row, and saving an edited row refuses the edit, while a shown column breaks its own rule. The rules are Required (blank, whitespace, empty rich text or an empty lookup object), the date rules below, min/max length, pattern, and email format. One message names every problem, e.g. "Please fill: Corrective Actions, Target Date · Target Date: past dates are not allowed". Auto-ID, formula, read-only and hidden or condition-hidden columns are never demanded. An edit save skips the date rules, so a row saved earlier keeps its old dates.; **Allowed Dates for table date columns.** Columns → Column Settings shows an "Allowed Dates" card under Default Date: Disable past dates, Disable future dates, Disable weekends, Earliest date, Latest date. It is saved on the column (`basic.disablePastDates`, `disableFutureDates`, `disableWeekends`, `minDate`, `maxDate`) and rides the seed's existing `defaultDate` slot, so no seed change is needed. A Date field dragged into the table keeps its own picker settings (`advanced.datepicker.*`); those apply when the column has no Allowed Dates of its own.; **Date cell.** The table's date cell (shared by definition and transaction tables) sets `min`/`max`, so the browser calendar greys the out-of-range days. A date that still breaks a rule (typed by hand, or a weekend) is refused with a message.; **Required marker.** A definition-table header shows a red `*` after the label of every column "+" demands (`isUserRequiredColumn`: Required and filled by the user, so an auto-ID column gets none). It sits outside the truncating label so a narrow column never clips it, in the builder canvas and at runtime alike.; **Shared logic.** All of it lives in `helpers/tableRowChecks.ts`. Length and pattern are read through `buildOverlayFieldConfig`, the same lookup the table's text editors use.	Required on a table column was never enforced when adding a row. The add path only asked "is any cell filled", and the auto-generated Task ID always is, so RCF's "Long Term Corrective action" took rows holding nothing but T00001. Table date columns had no allowed-date rules at all: the cell was a bare `<input type="date">`. The Submit-time "Check Required columns" toggle exists in code but is missing from the live component-properties seed, so it cannot be switched on.

Feature ID	Feature Name	Description	Impact
	User Manual: Download PDF	The Preview tab has a "Download PDF" button. It calls; `userManualApi.downloadPdf` (arraybuffer, 120s) and saves the server-rendered PDF under the; file name the server sends. Before downloading, it saves any unsaved Review & Customize; edits (`onBeforeDownload` → `persist("preview")`) so the PDF matches the preview. The; backend error message is decoded from the arraybuffer body. The HTML download and Print; buttons remain.	
	New chart type: Funnel	Added **Funnel** as a chart type, a twelfth tile in the Chart Type picker; (Chart Properties) of the generic **Charts** component. A funnel uses the same data; mapping as a pie: a **Stage Field**, plus a value (Count / Percentage / Sum / Average; or a value formula). New **Funnel Options**:; **Stage order:** largest first (default), smallest first (pyramid), data order, or; a custom comma/new-line list. A listed stage with no records shows as an empty 0; stage; unlisted stages follow at the bottom.; **Conversion %:** of the first stage (default), of the previous stage, or hidden.; A pyramid measures from its bottom (entry) stage.; **Stage labels:** inside / right / left / hidden.; **Gap** between stages (0–20 px).	Requested chart type for stage-by-stage drop-off (e.g. RCF stages).

4. Enhancements / Improvements

Enhancement ID	Module	Description	Benefit
	transaction-parameters	new Access property **Record Lock Rules** (`recordLockRules`, JSON, registered in; `parameterMetadata.js`): `[{ "roles": ["<role id>"], "values": ["OWNER", "REVIEW"] }]`. A caller is; locked when a rule for one of their roles lists the record's current Record Lock Field value — each; role is checked only against its own rule. Roles without a rule, and callers with no roles; (workflow updates, workers), are never held. Works alongside the single Record Lock Value / Roles; rule (either one locking is enough), so existing settings behave exactly as before.; `recordAccessPolicy.js` adds `parseLockRules` (unreadable or incomplete rules are dropped — a bad; row can only switch a rule off) and `ruleLocksCaller`; `evaluateRecordAccess` refuses with; `RECORD_ACCESS_LOCKED` and "This record is read-only for your role at its current stage."; Create-lock (`evaluateRecordCreate`) is unchanged.	the single lock rule could hold only one set of phases, already used for the Initiator; (RCF bugs 4 / 25). The COO needed the same "edit only on my turn" behaviour with different phases:; after Validate, the COO could still edit and still saw Validate / COO Reject (RF00006).
	data-archival-permissions	The three submenus (Data Archival, Data Archival Approval, Data Archival Execution); are now children of `submenu_009_data_mgmt` (Workspace → Data Management) instead of direct; Workspace children — in `menuData.json` and in what the migration writes:; `menu_access_details`: added under Data Management; an earlier run's Workspace-level entries; are moved there (routes merged).; Role trees: nodes are added under the role's Data Management node (admin roles full, others; disabled); Workspace-level nodes from an earlier run are moved there keeping their access.; A role with no Data Management node is left untouched (new counter `noDataManagement`).; Gateway route map, role flattening and the frontend permission lookup all walk nested; submenus, so nothing else changes. Still DRY-RUN by default; nothing applied to any server yet.	User request 2026-10-21 — "add Data Archival under Data Management".

Enhancement ID	Module	Description	Benefit
	data-archival-permissions	Shape is now Workspace → Data Management → Data Archival → (Data Archival Approval,; Data Archival Execution), in `menuData.json` and in what the migration writes. The migration was; rewritten around one placement table (`PARENT_OF`): every run puts each submenu under its; parent wherever an earlier run left it (moves keep access), works on a copy of each tree so a; dry run never mutates, and keeps the admin mobile-flag alignment. Access semantics unchanged; (each submenu still has its own grant; approvers / executors also need Data Archival view to; see jobs, which the nesting now reflects).	
	data-archival-permissions	Data Archival is ONE submenu under Workspace → Data Management. Its access types:; View = see jobs / settings / preview; Create = new job, preview, submit, cancel; Edit = edit a; draft / settings, approve, reject, execute, finalize, approve changed records.; `menuData.json`: Approval / Execution submenus removed; their 5 action routes added to Data; Archival's Edit list.; `authorise.middleware.js`: those 5 routes are POSTs (default action 'create'); a candidate; authorizes them against 'edit', like the existing bulk-delete / search special cases.; Migration: one submenu; removes Approval / Execution nodes left by earlier runs from the menu; and from roles, carrying any grant they gave into Data Archival (View → View, Create / Edit →; Edit) so no role loses access.; Separation of duties is unchanged: the master service still refuses an approver who created or; submitted the job (unless self-approval is switched on).	

Enhancement ID	Module	Description	Benefit
	user-manual	`buildOutline` now produces the full enterprise structure: Purpose; (+ Business Objectives / Problems Solved / Business Value), Introduction, Scope (+ Covered /; Not Covered), Target Audience, Prerequisites & Getting Started, Key Concepts, Navigation,; modules/features, Field Validations, Workflows / Process Flows / Reports (as configured),; Functional Impact, Business Scenarios, Best Practices, What's New, Troubleshooting, FAQs,; Conclusion, Glossary and Revision Recommendations. Each has a `sectionGuide` instruction.; `buildChunks` writes them in themed calls (`CHUNK_GROUPS`) so each response stays small.; The Field Validations call receives every feature's field names. Screenshots also go to the; Navigation call. `structuralFallback` gives fact-based content for each new section when; the AI is unavailable (never brackets). Tests: 15, including a no-placeholder check.	The Enterprise template showed `[List the business objectives…]` style guidance; because generation never wrote those sections.
	user-manual	`generate` loads the manual's template (Basic Information → Manual Template) with; `releaseTemplateFields.service.js#loadTemplateHtml`, the same loader Release Notes uses.; `draftUserManual` then narrows the outline with `filterOutlineForTemplate`: only sections whose; token the template uses (`TOKEN_SECTIONS`), plus module overviews for `{{configuration}}` and; feature sections for `{{step_by_step}}`. A sub-section whose parent isn't placed is promoted; to top level. No template, `{{manual_document}}`, or no section tokens → the full manual.; `generatedMeta.templateId` records the template used. Tests: 18.	Requested: the document should be generated based on the template selected on the; first page, instead of always generating every section.

Enhancement ID	Module	Description	Benefit
	user-manual	Prompts with screenshots list them numbered and ask the AI to put a; `[[screenshot:N]]` marker where each belongs inside the right section, with a 1–2 sentence; explanation in that section's `screenshots` list. `placeScreenshots` swaps markers for; `<img>` plus "Figure N — explanation" (numbered in reading order, each screenshot used once).; Screenshots the AI didn't place, or when the AI is off, are matched to a feature/module title; by caption or file name (longest title first), else go to Getting Started, after the; section's first paragraph. `sanitizeHtml` now keeps `<img>` with an http(s) src + alt only.; Tests: 22.	Requested: screenshots should sit in their respective place with an explanation,; not in a separate screenshots section.
	user-manual	The PDF no longer starts a new page before every numbered section, including the cover,; Document Control and Table of Contents. `MANUAL_PRINT_CSS` forces `break-before/after:auto` and; `min-height:0`. It overrides breaks and canvas heights whether they are saved inline or as the; template's own `#id` CSS rules. It replaces `dropBodyPageBreaks`, which only matched inline styles; on sections 4 and later. With the Enterprise template and the same content, the PDF went from; 23 pages to 5. Frontend: Print and Download PDF now sit in the Preview's top bar, next to the; template selector.	The downloaded PDF had large blank gaps between pages: the short cover table and the; one-row version table each took a full page.

Enhancement ID	Module	Description	Benefit
	definition-table	The Column Width box in Column Settings ("e.g. 150px, 20%...", saved on the column's `width` and `basic.width`) now sizes definition-table columns. A px width gives that column exactly that width. A % width is a share of the table, never below the column's usual minimum (160px, 140px when frozen, 220px for dynamic-field/attribute/value columns). A column with no width shares the leftover space as before. If every column has a width, the first column also takes any leftover space, so the action column stays at the right edge. Anything other than a positive px, % or bare number is ignored. The table scrolls sideways once the columns no longer fit. The new `definitionColumnTracks.ts` builds the tracks; header, input row, saved rows and footer already share one `gridTemplateColumns`.	Every column got `minmax(160px, 1fr)`, an equal share, and the Column Width box did nothing on this table type (transaction tables already honoured it). In RCF's "Long Term Corrective action" table inside the stepper, the long Corrective Actions text got the same 247px as Task ID and wrapped into very tall rows. Tables with no Column Width set render exactly as before.

Enhancement ID	Module	Description	Benefit
	documentation-hub	`documentationTemplates.ts` gains `enterpriseUserManual` (category "Enterprise", first in; Browse User Manual Templates). Modelled on the "Cavin Studio — Workflow Management End User; Manual" PDF: cover page with a details table, Document Control, Table of Contents,; Disclaimer, then numbered sections (Purpose, Introduction, Scope, Target Audience,; Prerequisites, Key Concepts, Functionalities, Process Flow, Navigation, Screen-by-Screen and; Step-by-Step, Field Validations, Workflows and Functional Impact, Business Scenarios, Best; Practices, Troubleshooting, FAQs, Conclusion, Glossary, Appendix Screenshots, Revision; Recommendations). Same look: navy headings, orange sub-headings, navy-header tables, page; breaks per section. Generated sections use the existing tokens. Sections the generator; doesn't produce yet carry grey bracketed guidance for the template author.	Requested: offer the enterprise manual format in the Browse User Manual Templates drawer.
	documentation-hub	The Enterprise template's guidance placeholders are replaced by new generated-section; tokens (`{{purpose}}`, `{{business_objectives}}`, `{{problems_solved}}`, `{{business_value}}`,; `{{scope}}`, `{{scope_covered}}`, `{{scope_not_covered}}`, `{{audience}}`, `{{key_concepts}}`,; `{{navigation}}`, `{{field_validations}}`, `{{functional_impact}}`, `{{business_scenarios}}`,; `{{best_practices}}`, `{{conclusion}}`, `{{glossary}}`, `{{revision_recommendations}}`). They are; mapped in `ManualPreviewTab` SECTION_TOKENS (anchored for outline navigation) and listed in the; template variables panel. The `entGuide` helper was removed.	
	documentation-hub	Review & Customize / Preview show an amber notice when `manual.templateId` differs; from `generatedMeta.templateId`. Generation is now template-specific, so the new template's; sections may not exist until Regenerate.	

Enhancement ID	Module	Description	Benefit
	documentation-hub	The Review document, the Preview standard layout, `{{manual_document}}` and the download; no longer append a Screenshots section (`buildScreenshotsHtml` removed); `{{screenshots}}`; renders empty. The Enterprise template drops "Appendix — Screenshots" (now 23 sections), and; saved templates have that heading removed by `legacyTemplateUpgrade.ts`. RichTextEditor gains; the Image extension (no base64) so editing a section keeps its screenshots. Inline images are; styled in Review, Preview and the editor. Content Input help text updated.	
	documentation-hub	The bottom toolbar of the Preview (zoom, "N sections", HTML download, Print) is; removed. Zoom moved next to "Download PDF", which replaces the HTML download and Print. The; preview frame now fills the page height. In the Enterprise template only the front matter; (Document Control, TOC) starts a new page. `legacyTemplateUpgrade.ts#dropBodyPageBreaks`; strips the forced break from sections 4+ in templates that were already saved.	The toolbar repeated the new button and left empty space. The PDF put every; section on its own mostly-blank page.

Enhancement ID	Module	Description	Benefit
	formula-orchestrator	`evaluate-form-batch` answers 200 with **`results` AND `errors`** — one formula; failing says nothing about the other nine, so the server reports both. Only `results` was ever; read. A formula that failed was therefore indistinguishable from one that produced nothing: the; field went blank, nothing reached the browser, and the field's own **Error Fallback Value** — read; only by `FormulaField`'s error handler, which this path never reaches — silently did nothing.; `webBatchEvaluator.ts` returns `{ results, errors }` instead of the bare results map.; `orchestrator.ts` — `FormulaBatchEvaluator` now accepts **either** shape, so the React; Native evaluator (which returns a bare map) needs no change; `readBatchResponse` normalises.; A failed target takes its node's `errorFallbackValue`, and nothing is written when none is; configured, so a form that never set one is byte-identical to before. New optional; `onErrors` reports each failure with the node it belongs to.; `graph.ts` — the node carries `errorFallbackValue`, read from `advanced`.; `reportBatchErrors.ts` (new) — shared by both orchestrator call sites. Deliberately **not** a; toast: a cross-feature lookup legitimately matches nothing while someone is still typing the; code it reads, so an interrupting message would be worse than the bug. `console.error`,; matching `FormulaField`'s own `console.error("Formula error:", err)`.	CADP QA run 2, clarification FB-060 (Q-04) — "lookup finds no matching record: both; fields stay blank, no error, toast or message". Confirmed on DEV: the server returned; `results: { field: null }` with `errors: ["❌ No records matched for: …"]` and the browser; discarded the errors. A documented setting that silently does nothing is the defect; blank on; no-match is not, and is unchanged where no fallback is configured.

Enhancement ID	Module	Description	Benefit
	rcf-uat-defects	Transaction Settings → Access gets **Record Lock Rules** (`recordLockRules`): one lock; per role, each with its own values of the Record Lock Field, stored as; `[{ "roles": ["<role id>"], "values": ["OWNER", "REVIEW"] }]`. A user is read-only when a rule for; their role lists the record's current value; roles without a rule are never held. Works alongside; the existing single rule (Record Lock Value / Roles).; `recordAccessPolicy.ts`: `parseLockRules`, `ruleLocksCaller`; `evaluateRecordAccess` returns; `RECORD_ACCESS_LOCKED` with "This record is read-only for your role at its current stage." The; list hides Edit and the record page opens View Only with Approve/Reject hidden — the existing; lock handling, so **no change to `PreviewPage.tsx`**.; Settings screen: `lock-rules` editor kind + `parseLockRuleDrafts` / `serializeLockRules`; (`paramRowFreeze.ts`), `LockRulesEditor` (`ParamValueCell.tsx`: a card per role with its; locked values, "Add a role…"), "ROLE: values · ROLE: values" display (`ParamValueDisplay.tsx`),; catalog label (`useParamConfigRules.ts`), Access bucket mapping (`transactionSettingsClosed.ts`).; The server enforces the same rule (master `recordAccessPolicy.js`, see its; `transaction-parameters.md`).	after the COO pressed Validate on RF00006, the COO could still edit RCF Details / COO; REVIEW and still saw Validate / COO Reject. The single lock rule was already taken by the; Initiator, with different phases.

5. Bug Fixes

Bug ID	Description
—	No fixed bugs recorded yet for this release.

6. Configuration Changes

Configuration	Description	Impact
Service-to-service calls use in-cluster Service addresses	In `.env.production`, the `*_SERVICE_URL` values for MASTER, CLIENT_WS, CLIENT_METADATA, NOTIFICATION, WORKFLOW, SEARCH, TEMPLATE, CHAT now point at the; in-cluster Services (`http://cadp-backend-<name>.cadp-dev.svc.cluster.local/<context-path>`); instead of `https://cadp.cavininfotech.com/<context-path>`. Browser-facing values (CORS, frontend,; approval and subdomain URLs) and external hosts are unchanged. `BACKUP_SERVICE_URL` stays public: no k8s Service for it in the repo.	Calls between services in the `cadp-dev` namespace were leaving the cluster through the; public load balancer and nginx ingress with a TLS handshake each time. Calling the ClusterIP; Service directly removes those hops. Ingress no longer applies to these calls, so its body-size; limit and 60s read timeout do not either; the app's own limits and axios timeouts apply.; If the `cadp-dev-env` k8s secret defines any of these keys, its value takes precedence.
Service-to-service calls use in-cluster Service addresses	In `.env.production`, the `*_SERVICE_URL` values for MASTER, RBAC, CLIENT_WS, NOTIFICATION now point at the; in-cluster Services (`http://cadp-backend-<name>.cadp-dev.svc.cluster.local/<context-path>`); instead of `https://cadp.cavininfotech.com/<context-path>`. Browser-facing values (CORS, frontend,; approval and subdomain URLs) and external hosts are unchanged.	Calls between services in the `cadp-dev` namespace were leaving the cluster through the; public load balancer and nginx ingress with a TLS handshake each time. Calling the ClusterIP; Service directly removes those hops. Ingress no longer applies to these calls, so its body-size; limit and 60s read timeout do not either; the app's own limits and axios timeouts apply.; If the `cadp-dev-env` k8s secret defines any of these keys, its value takes precedence.

7. API Changes

API Name	Type	Description
None	N/A	No API changes are included in this release.

8. Known Issues

Issue ID	Description	Workaround
	Role Management Change Effect	Because some menu structures have changed, certain features may return a 403 status code. To resolve this, edit the role, select the platform feature, and save the role again.
	Scheduler with Workflow Not Working	Use the conventional feature-wise approach wherever workflow scheduling is required.
	Formula Builder UI – Field Insertion Through Typing	This works in current fields. For external feature fields, it won't work, as there could be fields with same names from multiple features.
	Formula Builder – AI-Based Formula Writing/Editing Not Implemented	Use manual formula entry instead.
364	Organization Details → Basic Info: Website URL accepts invalid URL formats without validation.	Expected: Validate the URL format and show an error message for invalid values, such as www.example or example@123.www.example
	Builder Tool Not Available on Mobile Web	The builder tool is not supported on mobile web. This is implemented as expected — builder functionality is restricted to desktop browsers by design. It should be available on Tab though.
	Mobile – Formula Field Not Working in Tables & Charts	Formula fields are not evaluated correctly when used inside Tables and Charts on mobile. This is under review.

Issue ID	Description	Workaround
	Batch import now resolves lookup fields to record snapshots — The batch import pipeline (`POST /:featureId/data/batch-import/upload`, used by the; Import dialog) stored every lookup field as the raw display text from the sheet, e.g.; `"Hosur Town East"`, instead of the; `{ _lookupRecordId, valueFields, displayFields }` snapshot a manual form submit writes.; `batch-import.service.js` resolved reference *tables* but never called; `lookupImportResolutionService`, which is what the non-batch import path uses to build that; snapshot. Added `resolveLookupsForBatch()`, built the lookup source context once per job in; `buildRowContext`, and called it in `processOneBatch` between reference-table resolution and; auto-ID assignment.	A record imported this way renders correctly but carries no link to its source record,; so every `_lookupRecordId` reader — formula aggregates (`formulaAggregate.repository.js`),; report tables (`report-table.repository.js`) and distributor/tenant data scoping — silently; matches nothing. The defect is invisible on screen, which is what makes it dangerous: it was; found only by reading back 2,000 imported Outlet Master rows and checking the stored shape.; Confirmed present in existing bulk-imported data in the CK Bakery workspace.
	Revived the batch-create import test suite — `feature-data-import-batched-create.test.js` mocked `mongoose` without; `Schema.Types`. Once `transaction-setting.model.js` entered the dependency graph it read; `Schema.Types.Mixed` at import time, the mock threw, and the suite died before a single test; ran — reported as one failed suite with 0 tests. Added `Schema.Types` to the mock.	The whole batch-import path had no running test coverage and nobody could see it: the; suite looked like one red suite rather than 19 missing tests.

9. What to Test

- Login – Login, Sign In with Google (have to be configured in external providers in Tenants of metadata), Sign In with Microsoft (have to be configured in external providers in Tenants of metadata)
- **Create Application**
  1. Start from Scratch
  2. Choose/Create Vertical Template

- **AI-Assisted Creation & Ask FAQ** (Cavin's AI assistant — the AI launcher icon)
    1. **AI App Creation**
        a. From the workspace (any page other than Modules Management / Features Management / Builder Tool), open Cavin's AI and select the **Create** tab.
        b. Verify the helper line reads "Create builds a new application from your description".
        c. Describe the application to build and send. Verify creation progress is shown and "Application created from AI." appears on completion.
        d. Verify the generated application, its modules and features are listed and can be opened in the workspace.
        e. If the AI asks clarifying questions, answer them and verify creation continues (no success message is shown until something is actually created).
    2. **AI Module Creation**
        a. Open an application's **Modules Management** page, open Cavin's AI and select the **Create** tab.
        b. Verify the helper line reads "Create builds a new module here".
        c. Describe the module to build and send. Verify "Created N module(s) from AI." appears and the module list refreshes with the new module(s).
        d. Open the created module and verify it can be edited further.
    3. **AI Feature Creation**
        a. Open a module's **Features Management** page (or the Builder Tool), open Cavin's AI and select the **Create** tab.
        b. Verify the helper line reads "Create builds a new feature here".
        c. Describe the feature to build and send. Verify "Created N feature(s) from AI." appears and the feature list refreshes.
        d. From the Builder Tool, verify you are taken to the first created feature in the builder.
        e. Verify the generated form has appropriate components/fields, then publish it and verify records can be created, viewed and edited.
    4. **Create / Ask mode behaviour**
        a. In the Create tab, type a question instead of a build request. Verify nothing is created and an "Ask this in Ask mode" suggestion is offered; clicking it moves the question to the Ask tab.
        b. With a user who has only AI Builder or only AI Ask access, verify only that tab is available; with no create access, verify past conversations are read-only.
        c. Verify conversation history can be searched and reopened for each mode.

5. **AI Ask FAQ**

   Scope: AI Ask FAQ is trained only on the following platform documentation. Questions must be asked on these topics only:

   - CADP User Login Guide
   - CADP Functional Specifications
   - CADP Erudite Management Complete
   - CADP Workspace Guide
   - CADP User Preferences Guide
   - CADP Role Management
   - AI FAQ Guide Transactions Runtime
   - CADP Login Complete
   - CADP Erudite Management Complete Guide
   - CADP Features Guide
   - CADP Modules Guide
   - CADP Platform Architecture Report
   - CADP Templates Integrations Workflows Basics Guide
   - CADP Features And Submodules Guide
   - CADP Organization Guide
   - CADP Builder Tool Guide

   a. Open Cavin's AI, select the **Ask** tab and ask a question about platform configuration on one of the trained topics above.

   b. Verify a relevant and accurate answer is returned.

   c. Ask a follow-up or ambiguous question and verify it is handled gracefully.

   d. Ask a question outside the trained topics and verify AI Ask FAQ declines or returns an appropriate out-of-scope fallback rather than a fabricated answer.

   e. Verify Ask mode changes nothing in the workspace (it only answers).

- **User Management**

  1. Create/Manage Users
  2. Create/Manage Groups
  3. Import/Export Users
  4. Archive/Unarchive Users
  5. Locked Users
  6. Waiting Approval
  7. Please update existing users in each workspace.
  8. In the Users tab, when creating a new user, the Approval Type default is No Approval.

- **User Preferences**
    1. Profile update – first name, last name updating, profile picture updating.
    2. Change password
    3. Locale and Date-time updating
    4. Interface
        i. Theme change, color change. Color change here will override the brand color for that specific user of that workspace only.
        ii. Records per page is reflecting.
        iii. Language (currently only English accepted)
    5. Customization – module and features are visible based on visibility setting, quick access for that enabled feature is redirecting to feature transactions (records) page.
    6. Privacy & Security
    7. Miscellaneous – Default landing page, preferred currency, auto-save drafts, save interval
    8. Preferred user workspace
    9. Reset Preferences
- **Groups**
    1. Archive/Unarchive

- **Roles**
    1. Platform Roles
    2. Archive/Unarchive
    3. Workspace Role
    4. Feature Access Level
    5. Data Access Level
    6. User or Group Role Mapping
- **Workspace Management**
    1. Create Workspace
    2. Clone Workspace
    3. Archive Workspace
    4. Workspace Settings
- **Organization Management**
    1. Organization Details
    2. Organization Address
    3. Branding Details
- **Module/Feature Management**
    1. Create/Manage Module
    2. Create/Manage Features
    3. Design components for features
    4. Features and Modules can be published
    5. Audit Logs (Platform & Workspace)
    6. Version Logs (Workspace & Workflow)
    7. Staging
    8. Feature Card and Module Card Customization
- **Notification & Alerts**
    1. Email Notification & Templates – Create template and use template to send notification
    2. SMS Notification & Templates – Create template and use template to send notification
    3. WhatsApp Notification & Templates – Create template and use template to send notification
    4. In-App Notification & Templates – Create template and use template to send notification
    5. Push Notification & Templates – Create template and use template to send notification

- **Erudite Management**
  1. **Configuration (prerequisite)**
     a. Enable Erudite Management in Role Management before testing.
     b. Verify users without the role cannot access the Erudite Management module.
  2. **FAQs**
     a. Create a new FAQ with a question, answer, and category (Account, Billing, General, Technical).
     b. Verify the FAQ appears in the FAQ list after saving.
     c. Enable "Display this on the dashboard" and verify the FAQ is visible on the dashboard.
     d. Edit an existing FAQ and verify changes are saved correctly.
     e. Delete a FAQ and verify it is removed from the list.
     f. Search for a FAQ by keyword and verify correct results are returned.
  3. **Documentation**
     a. Upload a document (PDF or DOC, max 5 MB) with a name, description, and category.
     b. Verify the document appears in the Documentation list after saving.
     c. Attempt to upload a file exceeding 5 MB and verify a validation error is shown.
     d. Attempt to upload an unsupported file format and verify it is rejected.
     e. Enable "Display this on the dashboard" and verify the document is visible on the dashboard.
     f. Edit document details and verify changes are saved correctly.
     g. Delete a document and verify it is removed from the list.
  4. **Tutorial Videos**
     a. Upload a video (MP4, MOV, or MKV, max 50 MB) with a name, description, and category.
     b. Verify the video appears in the Tutorial Videos list after saving.
     c. Attempt to upload a file exceeding 50 MB and verify a validation error is shown.
     d. Attempt to upload an unsupported file format and verify it is rejected.
     e. Enable "Display this on the dashboard" and verify the video is visible on the dashboard.
     f. Edit video details and verify changes are saved correctly.
     g. Delete a video and verify it is removed from the list.
  5. **End User Access**
     a. Verify end users can browse and read FAQs, documentation, and tutorial videos.
     b. Verify items marked "Display on dashboard" appear on the user's dashboard.
     c. Verify items not marked for dashboard display do not appear on the dashboard.

- **Components**
    1. Layouts
        1. Column
        2. Section (various nested scenarios)
    2. Data Management (API Integration, Device Integration)
    3. Input Components (Phone Number, Password, Dropdown with Feature as Field Value)
    4. Specialized Input & Display
    5. Display and Interaction
    6. Media and Files
    7. Forms and Validation
    8. Utility Components
    9. Data Config Component
    10. Hyper Link Component
    11. Favorite Component
    12. OCR Component
        a. Add the OCR component to a feature form and upload an image or document.
        b. General text extraction → verify the text is extracted from the uploaded image/document and displayed within the selected output textarea
        c. Verify the extracted text is saved correctly with the record and is visible on View/Edit.
    13. Analytics & Report Components
        1. Chart Component
        2. Gantt Chart Component
        3. KPI Card Component
    14. Form UI
        1. Table component and Table level formulas, Summary row formulas
    15. Reusable Component
- **Page Builder**
    1. Create a new page and verify it is saved and listed.
    2. Drag-and-drop layouts (columns, sections) and reusable components onto the page canvas.
    3. Configure component properties and verify changes reflect on the page.
    4. Reorder, move, and delete components on the canvas.
    5. Edit an existing page and verify changes are persisted.
    6. Delete a page and verify it is removed from the list.
    7. Publish the page and verify it renders correctly at runtime.
    8. Verify the page is responsive across mobile, tablet, and desktop.
    9. Verify role-based access — only authorized users can build/edit pages.

- **Workflow Management**
    1. Visual canvas for creating workflows
    2. Drag-and-drop node placement
    3. Node connection with visual links
    4. Canvas zoom and pan controls
    5. Grid snap for alignment
    6. Create new workflows
    7. Edit existing workflows
    8. Delete workflows
    9. List/view all workflows
- **Workflow Nodes – Control Flow**
    1. Start Node (workflow trigger point)
    2. End Node (workflow termination)
    3. Condition/Decision Node (if-else logic)
- **Workflow Nodes – Communication**
    1. Email Notification Node
    2. SMS Notification Node
    3. User Task/Approval Node
    4. System Notification Node
- **Workflow Monitoring**
    1. Execution history logs
    2. Workflow instance status tracking
    3. Node-level execution details
    4. Error logs and stack traces
    5. Approval node changes for user selection
    6. Added SLA and escalation reminder for approvals
    7. Added parallel workflow approval node
- **Printing Templates**
    1. QR Code – Generate and Print
    2. Barcode – Generate and Print
    3. Invoice – Generate and Print
- **Role-Based Dashboards**
    1. Role-based dashboard template creation
    2. Dashboard rendering based on created template

- **Scheduler & Task Scheduler**
    1. Create and use Schedule
    2. Create and use Task Schedule
    3. Event-based task schedule
    4. Schedule-based task schedule
    5. Schedule a notification
    6. SMS
    7. Mail
    8. WhatsApp
    9. Push
    10. In-App
    11. Schedule a report send (PDF) through mail from a configured email template.
- **Formula Builder Functionality**
    1. Same Feature
        a. Arithmetic
        b. Logical
        c. Comparison
        d. Text
        e. Math
        f. Date
        g. Conditional
        h. Miscellaneous
        i. Aggregate → with filter conditions and without filter conditions.
    2. Cross Feature
        a. Arithmetic
        b. Logical
        c. Comparison
        d. Text
        e. Math
        f. Date
        g. Conditional
        h. Miscellaneous
        i. Aggregate → with filter conditions and without filter conditions.
    3. Incompatible datatype operations, for example, arithmetic on a text input field, should show a validation error.
    4. Unrecognized fields, functions, or operators should show a validation error.
    5. If external fields are called without where() conditions, it should show MULTIPLE RECORDS FOUND ERROR in preview for non-aggregate functions/operations.
    6. Variables are not working. Review which variables should be displayed in the modal.

- **Formula Builder UI**
  1. On clicking a feature from the feature list, fields in the selected feature should be displayed, except for the current field, to prevent self-call loops.
  2. On selecting external feature fields, where() should appear in the modal.
  3. On selecting where(), the modal should automatically switch to the Feature and Fields tab.
  4. On hovering over a field, field information such as module name and feature name should be displayed.
  5. The Clear button or backspace should work as follows:
     i. First click of clear/backspace will select the token.
     ii. Next click of clear/backspace will clear that token.
  6. Empty formulas should not be saved.
  7. Formulas should not be saved without validation.

- **Formula Builder – Helper Tools**
  1. **Cheat Sheet** (**?** icon)
     a. Open the Cheat Sheet and verify functions are grouped by category (Lookup, Nested Lookup, Aggregation, Conditional, Math, String, Date).
     b. Search for a function and verify the list filters correctly.
     c. Click the copy icon on an example and verify it is copied to the clipboard.
     d. Press Esc and verify the modal closes.
  2. **Examples**
     a. Open the Examples drawer and verify 50 scenarios across all categories are listed.
     b. Filter by a category and search; verify results update.
     c. Click "Use This Pattern" and verify the formula is inserted into the editor.
  3. **Test Formula (Playground)**
     a. For a current-feature calculation (e.g. *{number_field} * 5*), open Test Formula, verify a type-aware input appears per field (number/date/true-false/text), enter values, run, and verify the correct result, type, and (where relevant) matched-record count.
     b. For a lookup formula (uses *.where()* / external feature fields), verify NO value-input form is shown and a "Lookup formula – test against a real record" message appears instead.
     c. For an aggregate formula (SUM/COUNT/AVG/MIN/MAX of records), verify the "Aggregate formula – test against a real record" message is shown and the form is not available.
     d. Verify the Test Formula button is disabled when the formula is empty.
  4. **Template Library**
     a. Open the Template Library and verify the **Most Used** tab lists the 20 platform built-in templates (e.g. GST, TDS, net salary, lookups, aggregations).
     b. Search and filter by category; verify results update.
     c. Click "Use This Template" and verify the formula is inserted into the editor.
     d. With a formula in the editor, click "Save current formula as template", fill title/category/description, save, and verify it appears under **My Templates**.
     e. Delete a saved (non built-in) template and verify it is removed. Verify built-in templates cannot be deleted.
  5. **Formula Wizard**
     a. **Lookup mode:** choose "Lookup a value", select a source feature, select the field to return, add a match condition (verify the "Right side is a field reference" option shows a dropdown of the current feature's fields), check the live preview, and Insert.
     b. **Aggregate mode:** choose "Aggregate records", select feature/field, optionally add one filter condition, choose the function (SUM/COUNT/AVG/MIN/MAX), and Insert.
     c. Verify the formula preview only appears after a return field is chosen, and that going Back keeps your previous selections.
     d. After inserting, hover over the tokens in the editor and verify the correct Module name and Feature name are shown for each (including the source feature for the looked-up field).

6. **Smart Validation Messages**
    a. Enter an invalid formula (e.g. unknown function, missing where(), wrong argument count) and click Validate Expression; verify the error includes a helpful suggestion and an example, not just a generic message.

- **Integrations**
    1. Step 1: Go to the Integration Type field and select "Third-Party Software".

       Expected Result: A new dropdown field called Select Type should appear.
    2. Step 2: Check the options inside the Select Type dropdown.

       Expected Result: It should contain options including "Third-Party API" and "Google Drive".
    3. Step 3: Choose "Third-Party API" from the dropdown.

       Expected Result: The configuration fields specific to the Third-Party API should be displayed.
    4. Step 4: Change the selection to "Google Drive".

       Expected Result: The system should dynamically display the Google Drive configuration fields: Client ID, Client Secret, and Refresh Token.

This build — additional test focus:

- Faster, fail-closed role filtering on the module list — All master → client-ws calls now go through one shared HTTP client with keep-alive; agents (`src/utils/clientWsHttp.js`) instead of the global `axios`, so role checks reuse open; sockets rather than paying a TCP/TLS handshake per call; read-only calls retry once on; `ECONNRESET`. The role-check calls (`/role/view/modules`, `/role/view/features`,; `/role/view/features/:moduleId/:featureId`) moved into `ClientWsConnector`.; `getModulesByWorkspace` now runs the feature-count aggregation, the role-filtered recount, the; offline scope and the module role check concurrently (`Promise.all`) instead of one after; another, and `buildViewModules` posts identity stubs (id, tags, feature ids) and matches the; result back to the full documents instead of posting whole module documents.; *Behaviour change:** when the module role check cannot complete (client-ws down, timeout,; malformed response), the request now fails with 503 instead of returning the unfiltered; module list; an upstream 401/403 keeps its status. The role-filtered feature *counts* still; fall back to raw counts on failure.

10. What to Test on Mobile

- Login – Normal Login
- Login – SSO (Sign In with Google, Sign In with Microsoft)
- Workspace Switch
- Module List
- Feature List
- Components
- Feature Data – Create, View, Edit, Clone, Delete
- Dashboard
- Integration – Weighing Machine, Third-Party API (Budgie, SAP), and Payment Gateways (Stripe, Razorpay, and PayU)
- Theme
- Push Notification
- Workflow – Approve and Reject
- Charts – Feature Data
- Data Table – Feature Data
- Security – Prevent copy, paste, cut, and text selection